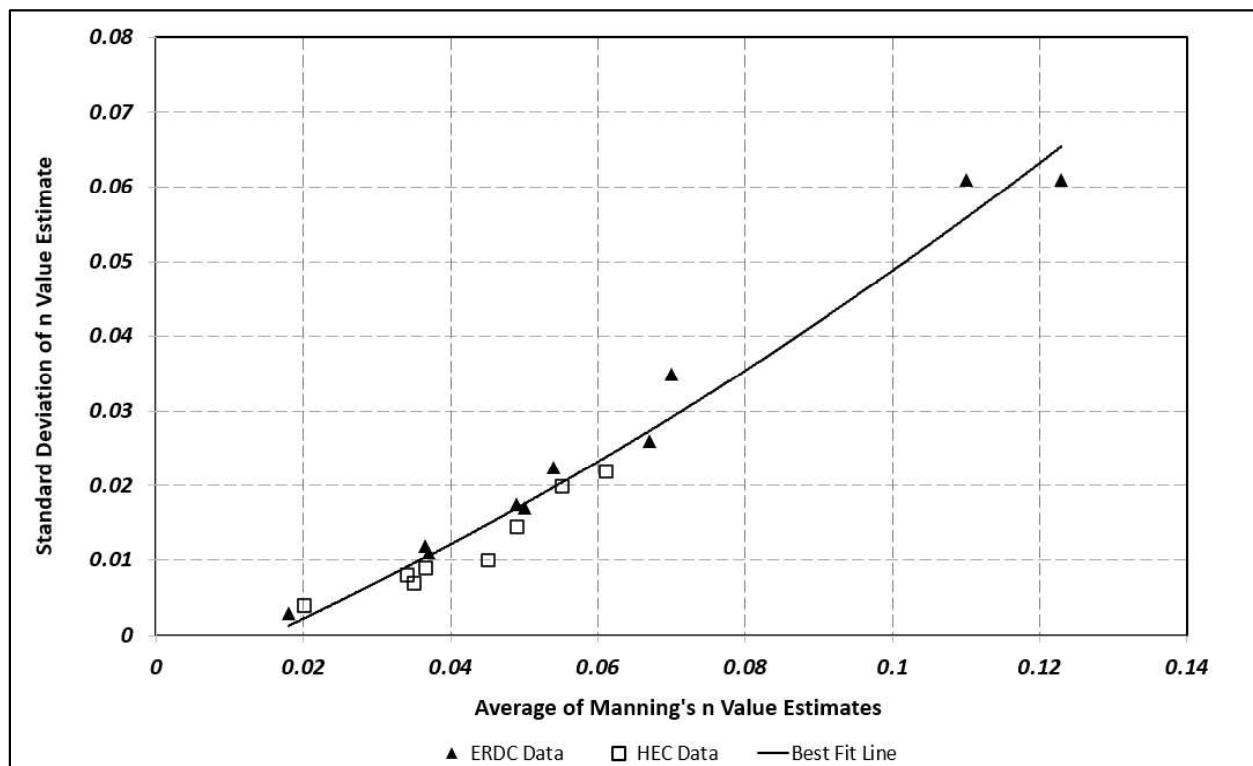


parameter (Manning's  $n$ ). An uncertainty distribution for the  $n$  values can be estimated through professional judgment or with published study results.

(c) An example of published results is shown in Figure 4–3, which displays results from HEC and USACE Engineer Research and Development Center (ERDC) studies, indicating the variation of uncertainty (standard deviation) about  $n$  values versus average  $n$  values. This plot can be used to help derive a reasonable probability distribution for Manning's  $n$  values to be used in Monte Carlo analysis, and sensitivity analyses, in the absence of more detailed site-specific evaluations.



**Figure 4–3. Uncertainty of Manning's  $n$  value estimates based on estimated mean values**

(3) Combining uncertainties for computed water surface profiles (stages from hydraulic modeling) involves the following:

(a) If the impact of factors that contribute to uncertainty can be characterized separately through statistical analyses, and if those factors are independent, then the combined uncertainty model can be described by simply adding the uncertainty variances. To obtain the total uncertainty in computed water surface elevations, equation 4–6 can be used.

$$S_{total} = \sqrt{S_m^2 + \cdots + S_n^2} \quad (4-6)$$

where:

- $S_{total}$  = the total standard deviation of uncertainty in stage
- $S_m$  = the standard deviation of uncertainty in stage due to the hydraulic model parameters
- $S_n$  = the standard deviation associated with other independent uncertainty sources
- $n$  = the number of sources of independent uncertainty included

(b) Clearly, as the uncertainty of each independent source increases and the number of independent sources increases, the overall uncertainty increases. Additional uncertainty sources, other than the hydraulic model parameters and the underlying terrain data, may include natural variability due to changes in bed forms, temperature, unsteady flow effects, seasonal roughness variations, changes in channel shape due to sedimentation, and many other factors. Any factor that contributes significantly to the uncertainty in the stage for a given discharge should be included in the uncertainty analysis. This may require separate analyses for that factor, then combining the uncertainties using equation 4–6.

(4) The preceding sections present a variety of methods for determining the standard deviation of uncertainty in stage derived using computed water surface profiles from hydraulic models. Whichever method is used, the standard deviation of uncertainty generated ( $S_{total}$ ) should be compared to Table 4–1, which shows reasonable minimum standard deviation values. Provision of minimum standard deviation values is an effort to provide consistency in USACE risk analyses and should be adopted when values computed in the study are lower than those minimums.

(a) The USACE research documents “Accuracy of Computed Water Surface Profiles” (HEC 1986) and Freeman, Copeland, and Cowan (1996) provide information for estimating uncertainty in water surface profiles obtained when using a gradually varied flow model. The standard deviation of the Normally distributed errors in the estimated stages are based on topographic information and confidence in estimated Manning’s  $n$  value, with minimum values shown in Table 4–1. Thus, the values shown in Table 4–1 reflect uncertainty in computed stages due to terrain and reliability of Manning’s  $n$  values. Given that numerical models usually use fixed terrain data, the uncertainty due to the terrain is not included in sensitivity or Monte Carlo analysis.

(b) Additionally, without an adequate number of observed events with which to calibrate, accurate awareness of the uncertainty in water surface elevations due to roughness coefficients may not exist, and typical analysis can underestimate these uncertainties. To ensure these factors are included, the values of standard deviation in stage errors should not be less than the minimum values shown in Table 4–1, unless there is sufficient modeling analysis and justification.

(c) The values in Table 4–1 represent standard deviations of uncertainty in stage at design-level discharges. Standard deviations for events less than the design level decrease as a zero discharge is approached. Standard deviations for events greater than design levels may be equal to or higher than what is shown in Table 4–1.

**Table 4–1**  
**Minimum standard deviation of error in stage**

| Standard Deviation (in feet)               |                                       |                                      |
|--------------------------------------------|---------------------------------------|--------------------------------------|
| Manning's n Value Reliability <sup>1</sup> | High-Resolution Geometry <sup>2</sup> | Low-Resolution Geometry <sup>3</sup> |
| Good                                       | 0.3                                   | 0.6                                  |
| Fair                                       | 0.7                                   | 0.9                                  |
| Poor                                       | 1.3                                   | 1.5                                  |

Notes:

<sup>1</sup> Where “good” reliability of Manning's n value equates to either excellent or very good model adjustment/validation to a stream gage, a set of high-water marks in the watershed upstream of the project, and other data. “Fair” reliability relates to fair to good model adjustment/ validation for which some, but limited, high-water mark data are available. “Poor” reliability equates to poor model adjustment/validation or essentially no data for model adjustment/validation.

<sup>2</sup> High-resolution geometry equates to data collected in clear and unobscured areas with soundings of underwater cross sections and high channel width to digital elevation model (DEM) width ratio (ratio is greater than or equal to 10).

<sup>3</sup> Low-resolution geometry equates to data collected in obscured areas with estimated underwater cross sections and low channel width-to-DEM width ratio (ratio is less than 10).

*c. Probability density functions for describing stage uncertainty.* The description herein suggests use of the Normal distribution to model uncertainty about stages in the stage-discharge transform curve. Using the Normal distribution is a common practice for models of errors and uncertainties when large samples are not available for comprehensive testing or when little published research is available to suggest an alternative. However, other distributions of the stage uncertainty may be used if the choice is supported and if parameters of the distribution can be estimated with the data and information available.

(1) Normal distribution: A two-parameter probability distribution defined by the mean and standard deviation. This distribution forms a symmetrical bell-shaped curve applicable to many kinds of data sets where values are equally likely to be greater than or less than the mean (symmetrical) and is also called a Gaussian distribution. The range created by the mean plus and minus two standard deviations spans 95% of the probability.

(2) Triangular distribution: A three-parameter probability distribution defined by a minimum stage, a maximum stage, and a most likely stage (mode). This distribution can be used for describing uncertainty that is not symmetric about the mean.

(3) Log-Normal distribution: A two-parameter probability distribution defined by the mean and standard deviation of the log of the variable. This is a nonsymmetrical distribution applicable to many kinds of data sets where the majority (more than half) of

values are less than the mean, but values greater than the mean can be extreme, such as with streamflow data.

*d. Describing stage uncertainty for the full range of possible flows.* A stage-discharge relationship must be derived for the full range of possible flows. Points on the curve are usually derived for the following probability-based events: 50%, 20%, 10%, 4%, 2%, 1%, 0.5% and 0.2%. Therefore, uncertainty about the mean stage estimates must also be described for the full range of events. Uncertainty is zero at the zero discharge stage. Uncertainty tends to increase as discharges increase toward the bank-full stage (while flows are contained in the main channel). Above bank-full stage, uncertainty in stage may increase, decrease, or stay the same. Specifically, when flows enter a floodplain, uncertainty may decrease because the change in stage for a given change in discharge becomes much smaller as the cross section widens. The full range of uncertainties can be computed from either the gaged stage-discharge data, or with a hydraulic model, as described in paragraph 4–2.

#### **4–5. Analysis complexity**

*a.* While the majority of water surface profile analyses are within the capabilities of one-dimensional, steady-state, hydraulic models, for streams that have rapidly varying flows or are subject to tides, an unsteady flow analysis may be needed. Sand bed streams may require mobile boundary modeling. Complex flow fields in unusual floodplains or estuaries may require multi-dimensional flow analysis. In such cases, when a single discharge-stage function is needed for a curve-combination risk analysis approach, a stage-discharge rating for the highest stages proportional to flow conditions of interest should be developed. The uncertainty associated with the rating is interpreted from the analysis results.

*b.* Often, sensitivity analysis as discussed below is an appropriate approach to such determination. If it is not possible to develop a rating from the results, analysis dealing directly with a stage-frequency relationship is likely to be necessary. Another option is use of the event-sampling approach to risk analysis, described in paragraph 2–8, which allows events to be modeled directly in hydraulic models during the analysis, and no one-to-one relationship between stage and discharge needs to be developed.

#### **4–6. Sensitivity analysis and professional judgment**

*a.* One approach to estimating stage uncertainty that can always be used, and was described for hydraulic modeling in paragraph 4–4.b(1)(a), is to estimate the upper and lower bounds on stage for a given discharge and convert that stage range to the needed uncertainty distribution. For example, if stage uncertainty follows a symmetrical Normal distribution, 95% of the possible stage range is encompassed by stages that are two standard deviations above and below the best estimate. Professional judgment could thus be applied to estimate the “reasonable” upper and lower bounds of stage, and the standard deviation estimated as the total range divided by four to determine a standard deviation.

*b.* Sensitivity analysis in which reasonable likely combinations of upper and lower bounds of model parameter values are also used to obtain a range of predicted stages for a given discharge could augment or serve as an alternative to the range determined from professional judgment.

#### **4–7. Stage uncertainty for additional project conditions**

The discussion in this chapter focused on estimating stage uncertainty for the existing without-project condition. The stage uncertainty for three additional conditions must also be evaluated: (1) the future without-project condition; (2) the initial with-project condition; and (3) the future with-project condition. If the flow and conveyance characteristics are expected to be markedly different from the existing without-project condition for any of those three additional conditions, further analysis is appropriate to estimate stage uncertainty specific to the differing conditions. If flow conditions and conveyance are expected to remain similar to the existing without-project condition, then stage uncertainty may be taken to be the same or similar to the existing without-project condition.